

BETWEEN STOPS

An interactive sound installation where three independent participants shape their own voices within a temporary shared space.



Link: <https://vimeo.com/1225209958?share=copy&fl=sv&fe=ci>

INTRODUCTION



Fig. 1. Between Stops, installation view, 2026. Photograph by Mingu (@anooo0518)

This is an interactive sound installation composed of three suspended handles. Drawing on the familiar image of handrails in public transportation, the work temporarily creates a shared space for multiple participants without requiring cooperation. Each participant alters their own independent sound by grasping and pulling the handles—where the force, rhythm, and duration of movement affect the sound's state. When participants' actions briefly synchronize, the system may produce a unified acoustic transformation. The piece explores how multiple independent individuals coexist within a shared environment, perceive one another, and occasionally form fleeting connections.

CONCEPT AND BACKGROUND RESEARCH

1. Initial Inspiration

This project originated from my personal experience riding a mountain bus. As the crowded vehicle navigated sharp turns, passengers shifted their center of gravity, grabbed handrails, and constantly adjusted their postures to adapt to the motion of the bus—being influenced by the road conditions. I am interested in these subtle bodily responses during travel, as they transform the ordinary act of moving into an embodied experience. In my initial proposal, I temporarily described the body as "an expression of motion itself," understanding the passenger's body as a medium through which environmental forces become perceptible.

2. Initial Research & Research Question

An important early theoretical reference was Tim Ingold's *Lines: A Brief History*. Ingold regards 'wandering' as a vibrant, responsive movement, contrasting it with point-to-point transportation logic(Ingold, 2007, pp.73-77, pp.101). This gave me a way of thinking about what is lost when a journey is reduced to origin and destination. My initial question was whether the apparently flattened "in-between" of transport could be reopened through bodily sensation: could the small tensions, movements and vibrations that remain in the body reveal another form of experience inside this connecting line? This question formed the conceptual basis of the first proposal.

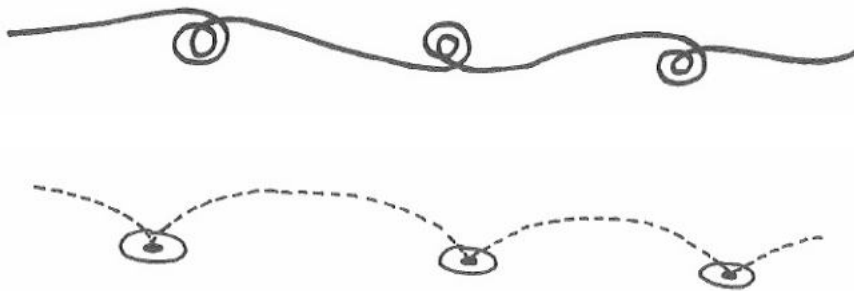


Fig. 2. Wayfaring (top) and destination-oriented transport (bottom). Reproduced from Ingold, T. (2007), *Lines: A Brief History*, p. 101.

However, the project gradually abandoned a literal reconstruction of the "bus journey." Public transportation, rather than being the subject of my project, functions more as a spatial and social reference within it. The suspended handrail is a familiar object associated with a shared environment in which strangers temporarily occupy the same structure.

For this reason, I use the transport handle as a shorthand for a public condition: it allows me to establish a temporary shared space quickly without constructing an entire transport environment. The work therefore does not attempt to simulate a bus, reproduce a specific route or tell a narrative about transportation. Instead, the visual language of public transport provides a framework in which several people can enter, act independently and remain aware of one another.

This shift also changed my understanding of ‘participation’. During the bus journey I initially mentioned, I observed that strangers sometimes let out nearly identical exclamations when the bus made a sharp turn, even though they spoke different languages. At first, I described this phenomenon as a brief resonance among strangers. My first proposal translated this observation into a much more explicit system: several participants would be subjected to a shared virtual force, and simultaneous tension peaks would briefly merge their individual sounds into a collective voice.

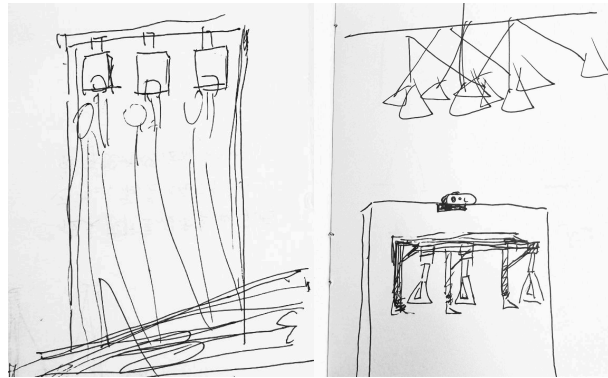


Fig. 3. Early concept sketches. Left: a physical mechanism designed to apply force back to the participant. Right: a camera-based system proposed to track the direction of handle movement, 2026.

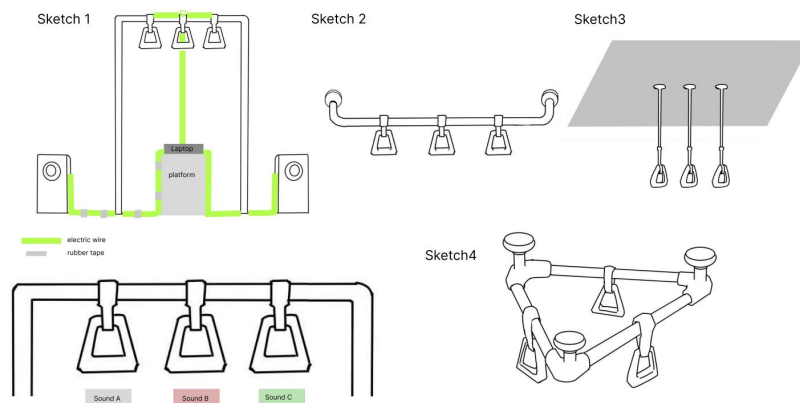


Fig. 4. Another Concept sketches, 2026.

As the project developed, however, I became less interested in reproducing synchronisation as a designed outcome. If synchronisation became something participants were expected to discover or achieve, the installation risked becoming a collaborative game with an implicit success condition. I instead wanted the work to remain valid when one person interacted alone, when two people ignored one another, or when three people acted independently. A relationship between participants could emerge, but it should not be required for the system to function.

3. Conceptual Framework

This developed position brought the project closer to questions of co-presence rather than cooperation. Once the physical bus movement was removed and participants were given direct control over the handles, the interaction changed. The participants were no longer only bodies responding to an external environment; their actions began to constitute the environment experienced by others.

Erving Goffman's *Behavior in Public Places* is useful for understanding this kind of situation. Goffman examines forms of interaction that take place between people who are physically co-present, including "unfocused interaction", where individuals share a situation without necessarily entering into direct communication (Goffman, 1963, pp. 17–24). This is close to the social condition I want the installation to create. The participants do not need to speak, cooperate or even consciously perform together; nevertheless, their movements and sounds become part of the same perceptual environment.

Edward T. Hall's concept of proxemics also provides a useful framework. In *The Hidden Dimension*, Hall describes interpersonal distance and the way spatial organisation influences social relations and bodily behaviour. The three handles create three temporary positions within the same structure, but these positions are not experienced identically. Height, reach, posture and habitual interpersonal distance influence how participants stand, how strongly they pull, and how closely they position themselves to others. Rather than treating these differences simply as noise within the sensing system, I became interested in allowing bodily difference to remain visible and audible.

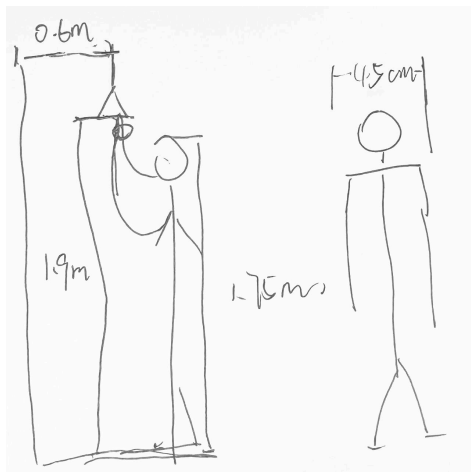


Fig. 5. Early measurements informing handle height and spacing. Author's sketch, 2026.

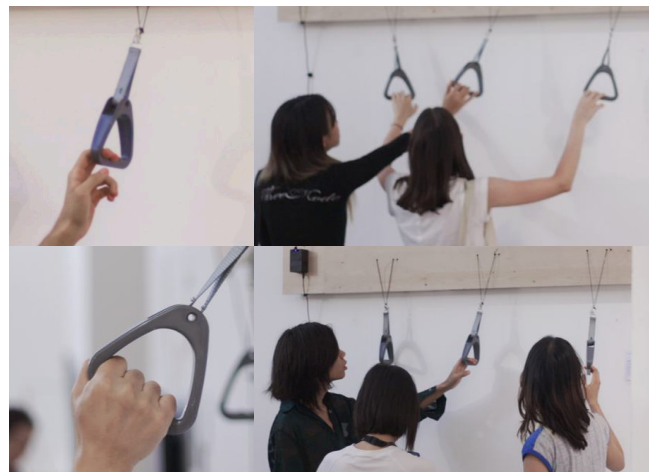


Fig. 6. Different gripping postures observed during the final exhibition. Video stills, 2026.

Sound supports this structure because it is simultaneously individual and spatial. Brandon LaBelle's writing on sound and everyday acoustic environments helped me think about listening as relational rather than purely representational. A sound can originate from one action while occupying a space shared with several people. Sounds can overlap, interfere and become perceptible without requiring participants to look directly at one another (Brandon LaBelle, 2019, pp. xi–xviii). In the installation, sound therefore operates not simply as feedback from a sensor, but as one of the means through which another participant becomes present.

Public transport remains important within this framework, but its role has also changed. I no longer consider the work to be primarily about a bus journey. Instead, the suspended transport handle acts as a familiar spatial cue. It quickly suggests a condition in which strangers temporarily occupy the same infrastructure, retain their individual positions, and are free to enter and leave. This allows me to construct a minimal form of public space without attempting to reproduce an entire vehicle.

4. Artistic And Computational Context

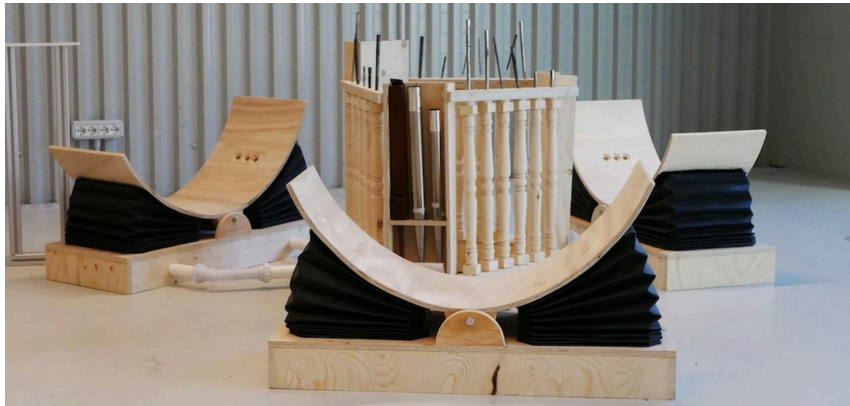


Fig. 7. Zixi Li, Rocking Pipe Organ. Image reproduced from UAL Graduate Showcase.

My initial research particularly focused on how synchronicity and bodily differences produce audible effects in the work, making *Rocking Pipe Organ* an important artistic reference during my early creative phase. This piece is made of three interconnected bellows chairs, requiring participants to coordinate their physical strength to produce harmonious sounds. But my installation work eventually took a different direction—I no longer pursued unity and harmony, yet retained the mechanism of 'accidental resonance.'

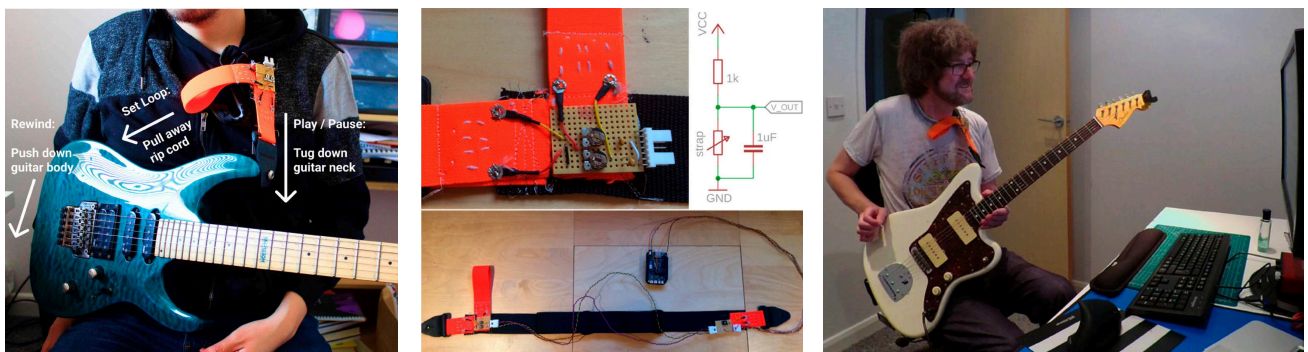


Fig. 8–10. Selected images from The Stretchy Strap. Reproduced from Martinez Avila et al. (2023).

At the same time, I had long planned to map stretch into data as an input, and Martinez Avila provided a precedent for this. His project *The Stretchy Strap* developed an elastic guitar strap that uses body and postural movements as inputs for interacting with digital media. Researchers constructed textile strain sensors using elastic fabric and conductive yarn; changes in resistance were read through a voltage divider circuit and sampled by a Bela embedded computer. At this point, I became familiar with the Bela board and began using it, and the initial outline of this project started to take shape.

In summary, this project integrates physical computing, interactive sound, and participatory space. Its evolution has shifted from an early interest in passively engaging with the environment through the body, toward a relational system in which body, object, sensing technology, and sound continuously influence one another, co-shaping each other's experiences.

TECHNICAL IMPLEMENTATION

1. Interaction Design And System Logic

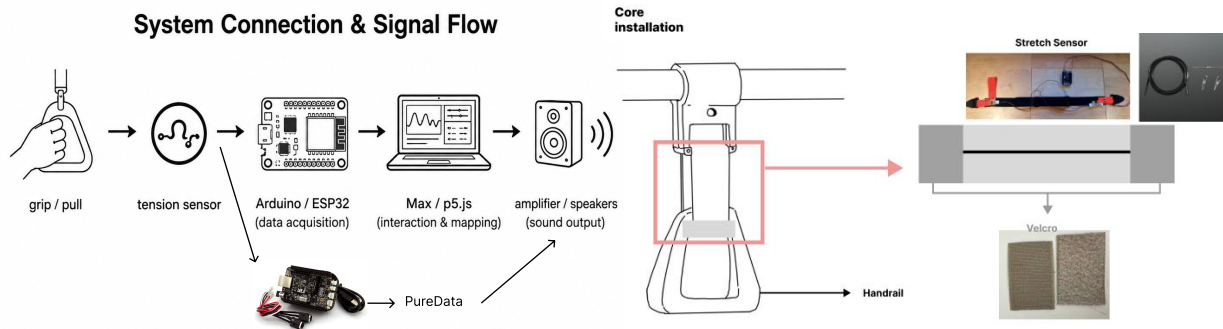


Fig. 11-12. Author's sketch, 2026.

This device uses familiar public object handles to suggest participant actions, eliminating the need to learn complex abstract interfaces. The interactive system employs simple physical interactions, as shown in the left image: participants grasp, pull, hold, and release any of the suspended handrails. When a handle is pulled, the tension in the sensing material changes, resulting in a corresponding change in resistance. The Bela board reads these analog values and converts them in real time via PureData into sound output through DIY speakers. Thus, the most critical component is the stretch sensor section, illustrated in the right image.

2. Interaction Design And System Logic



Fig. 13 Material experiments, 2026.

For the sensor component, I conducted material experiments. Initially, I attempted to manually craft textile strain sensors, drawing partial inspiration from Martinez Avila et al.'s *'The Stretchy Strap'* project. Subsequently, I tried crocheting my own stretch sensors. However, the resistance values proved unstable, and the material's elasticity was limited, making these handmade textile sensors unsuitable for installation requirements.

Therefore, I shifted direction and used stretch rubber cord instead to prototype the sensor.

3. Circuit Design And Hardware Development

After selecting the sensor, I designed a simple circuit to read the stretch-based input signal through the Bela board. Each sensor is connected to Bela's analog input through a voltage divider circuit using the sensor's built-in 10 kΩ resistor. Early testing was conducted on a breadboard, allowing for quick changes to connections and resistor values during the prototyping process.

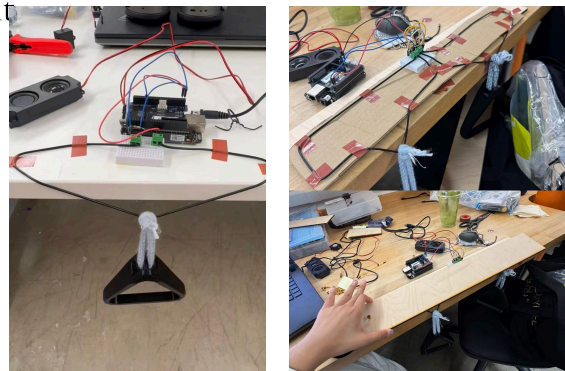


Fig. 14-15 Prototype 1, 2026.

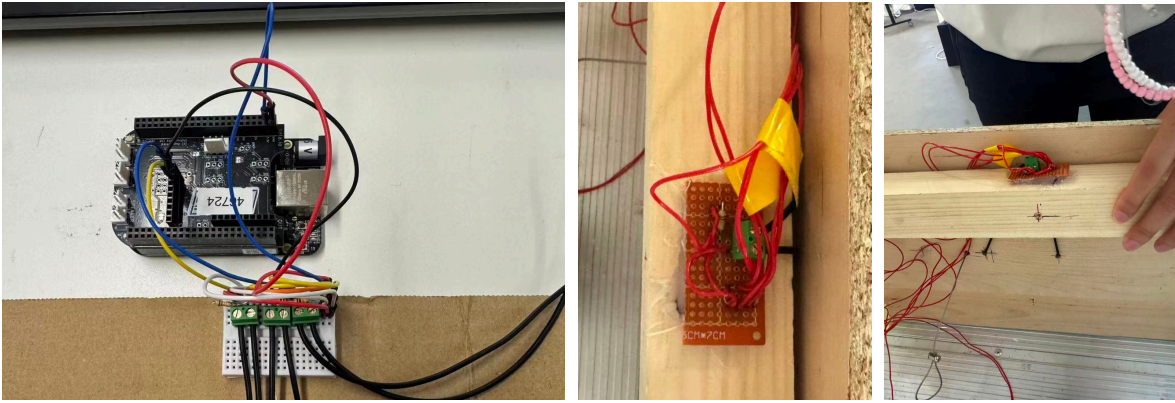


Fig. 16–18. Development of the circuit from breadboard prototype (Fig. 16) to soldered version (Figs. 17–18). Author's photographs, 2026.

The project subsequently expanded from a single-channel prototype to a three-channel system. At this stage, although the breadboard system is convenient for development, its stability is insufficient to meet the requirements of final installation. After stabilizing the circuit design, I re-soldered the connections on the circuit board to improve reliability and reduce the likelihood of loose connections during installation.

4. Sound Mapping In Pure Data

The mapping of resistance changes due to stretching and the sound synthesis were developed in Pure Data running on a Bela board, with each handrail assigned its own input channel. I first smoothed the sensor readings and then mapped them to sound parameters. Calibration for each sensor was performed experimentally, collecting data on resting value ranges, tension value ranges, and post-release value ranges. For example, for the first channel, I recorded resting values approximately between 0.500–0.507, tension values around 0.439–0.461, and post-release values roughly 0.487–0.496. Each physical sensor required individual calibration rather than using a shared, universal threshold.

During development I tested different approaches, including more layered and complex timbral behaviours. However, these often moved the work too far towards a descriptive or overly musical sound. In the final version, I chose to retain sine-wave-based sound generation. This created a simpler and more direct relationship between pulling force and sonic output, while also maintaining a coherent sonic language across the three handles.

This decision was also conceptual. I did not want the sound to overdetermine the interpretation of the work or become too illustrative. The relative sparseness of the sine-wave output allowed the sound to function more as a marker of bodily presence and tension within a shared space.

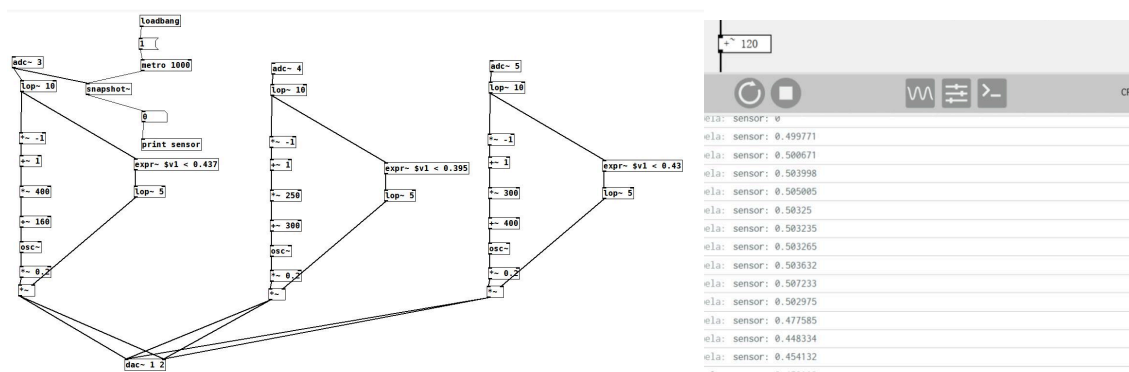


Fig. 19–20. Pure Data sound-mapping patch (Fig. 19) and sensor-value console used during calibration and testing (Fig. 20), 2026.

4. Installation Prototyping And Structural Iteration

Structural development proceeded in parallel with the development of electronic components. In addition to sensing and audio systems, I also developed several solutions for mounting handrails. Early design proposals primarily focused on load-bearing capacity and how multiple participants could stand in front of a single wall-mounted handrail. Some of these designs offered stronger structural support but created difficulties for cable routing and electronic component installation (Fig.21-23).

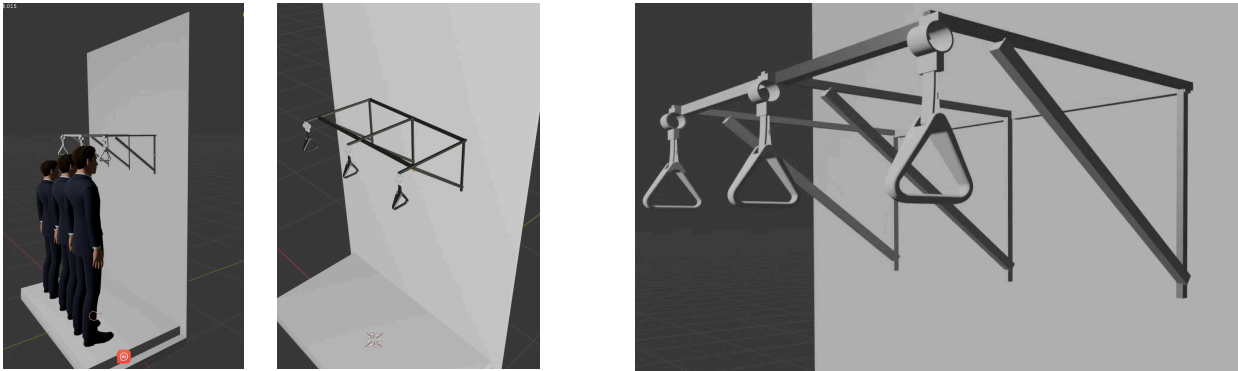


Fig. 21-23. Early installation models exploring wall-mounted support structures and participant positioning. Author's renderings, 2026.

After discussing installation options with Owen, I initially considered using a simple sandwich-clamping structure to secure the suspended handrail. At this stage, key considerations included load capacity, suspension height, and how the structure would accommodate cabling. I also used visual planning images—including AI-generated renderings of the installation—to test how the structure would appear in space and to explore possible participant positions in front of it. These images were not records of the final product but rather part of the design process, helping me think through scale, spacing, and audience arrangement.

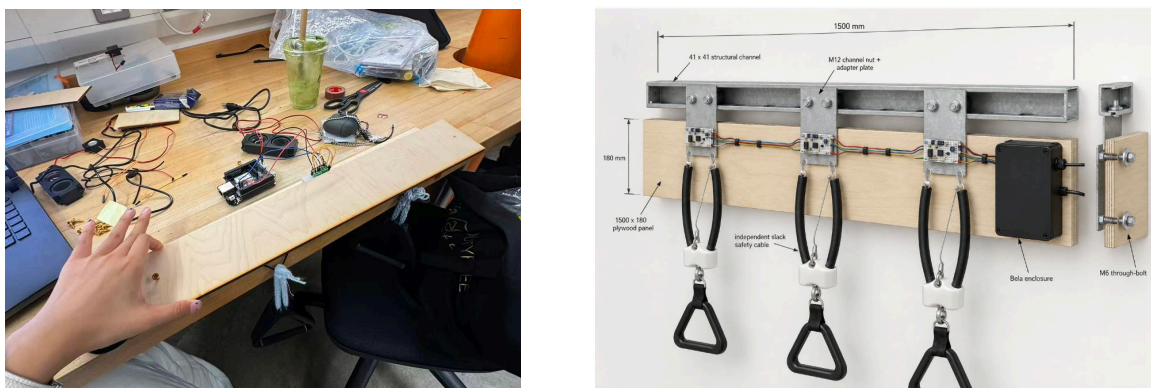


Fig. 24-25. Physical installation prototype (Fig. 24) and AI-generated installation visualisation used to explore mounting, scale and component arrangement (Fig. 25), 2026.

Once the three-hand system was assembled, I conducted wall-mounting tests. The results revealed another series of critical practical issues. During testing, one sensor broke under load, indicating that the sensing material and cable pathways remained too fragile for repeated use. To address this, I added additional mechanical protections: limiters were installed to prevent overextension, and heat-shrink tubing was inserted into drilled holes in the wooden structure to reduce wear at points where sensors passed through the wood.

REFLECTION

1. Reflection On The Process

This project became considerably more technically and materially demanding than I had initially expected. I entered the project with previous experience in Arduino and Max/MSP, but had not previously worked with e-textiles, Bela, Pure Data or sound installation. As a result, much of the development process involved learning new tools while simultaneously attempting to make them stable enough for exhibition.

One of the clearest lessons was the difference between a functioning prototype and a reliable installation. Early tests proved relatively quickly that stretching a sensor could be translated into sound, and I was able to run an early Bela prototype independently from the laptop. My journal records this transition from initial sensing tests to standalone operation and then to multi-sensor calibration. However, expanding the system to three physical interfaces introduced problems that were difficult to anticipate at prototype scale: different sensor ranges, signal drift, mechanical strain, cable abrasion, loose breadboard connections and eventually sensor breakage after wall-mounted testing.

These failures changed the way I approached making. Instead of treating technical problems as interruptions to the “real” artwork, I gradually understood them as part of the design process. Replacing unstable handmade textile sensors, soldering the final circuit, adding mechanical limiters and protecting the sensor where it passed through the wooden structure were all responses to what the material system revealed through use. The final form was therefore produced through repeated cycles of testing, failure and modification rather than through the execution of a fixed plan.

The main weakness of the final version is the sound design. I experimented with more complex mappings and timbres, but eventually returned to relatively simple sine-wave voices in order to prioritise interaction reliability. While this produced a clear distinction between the three handles, I do not yet feel that the sonic language has developed to the same degree as the physical and conceptual structure of the installation. This is an unresolved part of the work rather than a decision I consider fully complete.

2. What I Learned

The project significantly expanded my technical practice. My early research into e-textiles led me to experiment with constructing my own pressure and stretch sensors. Although these prototypes remained unstable, I gained a practical understanding of conductive materials, resistance-based sensing and the difficulties of designing deformable interfaces.

Working with Bela and Pure Data was another major development. My previous experience with Arduino and Max/MSP made the underlying logic familiar, allowing me to adapt relatively quickly, but this was my first time building a standalone real-time sensing and sound system with these tools. The project has left me particularly interested in the intersection between soft or textile interfaces and sound, which I would like to continue exploring beyond this project.

Working with Bela and Pure Data was another major development. My previous experience with Arduino and Max/MSP made the underlying logic familiar, allowing me to adapt relatively quickly, but this was my first time building a standalone real-time sensing and sound system with these tools. The project has left me particularly interested in the intersection between soft or textile interfaces and sound, which I would like to continue exploring beyond this project.

I also learned considerably more about physical installation. My initial plans underestimated the amount of time required for mounting, load testing, cable management and mechanical protection. Because several of the technologies were new to me, I should have reserved substantially more time for full-scale installation testing rather than treating it as the final stage after the electronics were working.

At the same time, the project made me recognise the value of communicating problems early. Regular updates to technicians, tutors and my supervisor often produced straightforward solutions to problems that had felt difficult when working alone. I also began keeping a development journal more consistently. Recording what had been achieved, what was still unresolved and what needed to happen the following day became an unexpectedly useful working method. The journal contains not only progress notes but sensor ranges, threshold decisions and installation problems—for example, recording different resting and active ranges for individual sensors rather than relying on memory. This reduced the cognitive load of returning to the project each day and made the development process easier to evaluate retrospectively.

3. Future Development

During my Viva process, Milly recommended David Byrne's **Playing the Building**. In this installation, an interface resembling an organ activates motors, percussive elements, and air systems directly connected to architectural components—such as metal beams, pipes, and ducts—effectively transforming the building itself into a sounding body. This resonates with Jeremie's recommendation of artist Jean Tinguely, both involving physical sound generation. These references are particularly relevant to an unresolved question in my project: rather than merely using electronic synthesis as feedback, could the physical structure of the installation itself generate or shape sound?

Future iterations could explore transducers, contact microphones, or mechanically excited materials so that pulling a handle causes wooden supports, metal fittings, or tensioned elements to resonate. This would create a tighter connection between sound and the physical forces originally driving the project, potentially moving beyond the current version's relatively abstract sine wave output.

I will also return to the relational dimension among the three participants. I don't necessarily need to reintroduce the initial concept of clearly defined "successful" synchronized events. Instead, I'm more interested in subtler forms of acoustic coupling: one participant's movement might slightly alter another's perception of resonance, spatial awareness, or acoustic conditions, while each person's voice remains distinct. In this way, future versions could further develop the conceptual shift emerging during the project—from understanding the body as a medium for environmental forces, to viewing multiple bodies, objects, and sounds as temporarily co-constituting an environment.

REFERENCES

Byrne, D. (2008) Playing the Building [interactive sound installation]. Battery Maritime Building, New York: Creative Time. Available at: <https://creativetime.org/projects/playing-the-building/> (Accessed: 9 September 2026).

Goffman, E. (1963) Behavior in public places : notes on the social organization of gatherings. New York: Free Press.

Hall, E.T. (1969) The hidden dimension. Garden City, N.Y: Doubleday.

Ingold, T. (2016) Lines A Brief History. 1st edition. London: Routledge

LaBelle, B. (2010) Acoustic Territories: Sound Culture and Everyday Life. New York: Continuum.

Li, Z. Rocking Pipe Organ. UAL Graduate Showcase. <https://ualshowcase.arts.ac.uk/project/714864/cover>

Martinez Avila, J., Hazzard, A., Greenhalgh, C., Benford, S. and McPherson, A. (2023) 'The Stretchy Strap: supporting encumbered interaction with guitars', Journal of New Music Research, 52(1), pp. 19–40. doi:10.1080/09298215.2023.2274832.

GENERATIVE AI DECLARATION

I acknowledge the use of ChatGPT (OpenAI, <https://chatgpt.com/>) during the development and documentation of this project.

1. Technical Learning, Debugging And Code Development

ChatGPT was used as a technical learning and debugging aid while I was developing the Bela and Pure Data system. I used it to help me understand Pure Data signal flow, sensor calibration, threshold logic, movement detection and audio mapping.

Representative prompts included:

- “I am using Pure Data and my patch currently looks like this. How should I connect it?”
- “Why does the sound continue after I release the sensor?”

The responses were used as explanations and debugging guidance while I manually constructed and tested the Pure Data patch. I adjusted the sensor thresholds using analogue values measured from my own physical sensors and repeatedly tested the system on Bela.

No AI-generated Pure Data patch was used as the final submitted implementation. The final patch was manually assembled and modified by me in Pure Data based on my understanding of the signal flow and the results of physical testing.

2. Writing And Documentation Support

I also used ChatGPT to assist with structuring and refining the written documentation for this project. I provided my own proposal, development journal, research papers, technical notes, photographs and reflections, and used ChatGPT to help organise these materials and improve the clarity of the academic writing.

Representative prompts included:

- “Help me reorganise the Concept and Background Research section.”
- “How should I describe the conceptual shift from environmental agency to relationships between participants?”
- “Help me organise the Technical Implementation using my journal and process documentation.”
- “Help me format this Link in Harvard style.”

The generated text was used as drafting and editing support. I reviewed, corrected and revised the outputs to reflect the actual conceptual and technical development of the project.